1. **Find the rank of the matrix**

a) $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$

b) $\mathrm{B} = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{bmatrix}$

Solution:

a) Subtract row 1 multiplied by 2 from row 2: $R_2 = R_2 - 2R_1$.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 3 & 5 & 7 \end{bmatrix}$$

Subtract row 1 multiplied by 3 from row 3: $R_3 = R_3 - 3R_1$.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix}$$

Multiply row 2 by -1 : $R_2 = -R_2$.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -1 & -2 \end{bmatrix}$$

Subtract row 2 multiplied by 2 from row 1 : $R_1 = R_1 - 2R_2$.

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & -1 & -2 \end{bmatrix}$$

Add row 2 to row 3: $R_3 = R_3 + R_2$.

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

So, the rank is 2.

b) Swap the rows 1 and 2 :

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 3 & 1 & 1 & 3 \end{bmatrix}$$

Subtract row 1 multiplied by 3 from row 3: $R_3 = R_3 - 3R_1$.

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & -5 & -8 & -3 \end{bmatrix}$$

Subtract row 2 multiplied by 2 from row 1 : $R_1 = R_1 - 2R_2$.

$$\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & -5 & -8 & -3 \end{bmatrix}$$

Add row 2 multiplied by 5 to row 3: $R_3 = R_3 + 5R_2$.

$$\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 2 & 2 \end{bmatrix}$$

Divide row 3 by 2: $R_3 = \frac{R_3}{2}$.

$$\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Add row 3 to row 1: $R_1 = R_1 + R_3$.

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Subtract row 3 multiplied by 2 from row 2: $R_2 = R_2 - 2R_3$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

So, the rank is 3.

5. Find a basis for each of the four fundamental subspaces associated with the matrix.

A= $\begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{0} \\ \mathbf{1} & \mathbf{2} & \mathbf{0} & \mathbf{1} \end{bmatrix}$

Solution:

Subtract row 1 from row 3: $R_3 = R_3 - R_1$. $\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Subtract row 2 multiplied by 2 from row 1: $R_1 = R_1 - 2R_2$. $\begin{bmatrix} 1 & 0 & -2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

So $rref\ (A) \ = \begin{bmatrix} 1 & 0 & -2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Colum Space: The column space is a space spanned by the columns of the initial matrix that correspond to the pivot columns of the reduced matrix.

Thus, the column space is $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$.

Row Space: The row is a space spanned by the nonzero rows of the reduced matrix.

Thus, the row space is $\left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \right\}$.

Null Space: Convert the matrix equation back to an equivalent system:

$$\begin{aligned} x_1 + -2x_3 + x_4 &= 0 \\ x_2 + x_3 &= 0 \end{aligned}$$

Add an equation for each free variable:

$$\begin{aligned} x_1 + -2x_3 + x_4 &= 0 \\ x_2 + x_3 &= 0 \\ x_3 &= x_3 \\ x_4 &= x_4 \end{aligned}$$

Collect terms into vectors: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2x_3 \\ -x_3 \\ x_3 \\ 0 \end{bmatrix} + \begin{bmatrix} -x_4 \\ 0 \\ 0 \\ x_4 \end{bmatrix}$

Factor out variables on the right side: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

Thus, a basis for the null space is: $\left\{\begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}\right\}$

Left Null Space: The left null space of a matrix A is basically the null space of A^T

So, we begin with determining A^T and arrange in the gauss elimination where the result is zero.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Add 2 times row 1 to row 3 : $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$

Add -1 times row 1 to row 4 : $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Add -1 times row 2 to row 3 : $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Convert the matrix equation back to an equivalent system: $\begin{aligned} x_1 &= 0 \\ x_2 &= 0 \end{aligned}$

Add an equation for each free variable: $\begin{aligned} x_1 \quad &= 0 \\ x_2 \quad &= 0 \\ = x_3 \quad &= x_3 \end{aligned}$

Solve for each variable in terms of the free variables: $\begin{aligned} x_1 \quad &= 0 \\ x_2 \quad &= 0 \\ x_3 \quad &= x_3 \end{aligned}$

Collect terms into vectors: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Thus, a basis for the null space is $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$